Snap-HyRE for Legal RAG: A Source-Gated Results Packet

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Abstract

This draft reports the current fixed-method Snap-HyRE evaluation package for legal retrieval-augmented generation. Snap-HyRE is a two-call generated-query method: first produce a tentative legal issue frame and hypothetical reasoning passage, retrieve with that passage, then answer from the original question and retrieved corpus evidence. The current signed package is incomplete (51/84 expected answer cells), so missing rows are left as TBD. In the signed full-corpus cells, Snap-HyRE improves BarExamQA answer accuracy over raw RAG for Gemma 4 26B (78.0% to 82.0%, $p = 0.000699$) and Llama 3.3 70B (74.6% to 79.8%, $p = 2.70{\times}10^{-5}$). On SCALR and CaseHOLD, generated retrieval often improves exact Hit@5 while final answer accuracy is flat or lower. HousingQA currently has raw and oracle rows but no signed generated-method answer rows. The main claim is therefore intentionally narrow: retrieval exposure and legal answer accuracy must be reported together, and Snap-HyRE is currently positive on BarExamQA, mixed on SCALR/CaseHOLD, and unfinished on HousingQA.

1 Draft Status

This is a sparse, source-gated paper draft. It treats the current generated package as the paper's unit of evidence and leaves missing cells visible. The active framing is a fixed-method Snap-HyRE evaluation, not a dataset-specific route-selection story. Result claims should continue to be checked against `docs/signoff_log.md`, `docs/compiled_results.md`, `logs/experiments.jsonl`, and the generated package before submission.

2 Method

2.1 Evaluation Contract

The method under test is fixed across datasets and models. We evaluate the same answer setting at $k = 5$ on BarExamQA, HousingQA, CaseHOLD, and LegalBench-SCALR. We compare seven rows: `llm_only`, `rag_simple`, `rag_hyde`, `snap_hyre`, `rag_rewrite`, `golden_passage`, and `golden_plus_neighbors`. The oracle rows are controls for evidence use, not deployable methods.

2.2 Fixed Snap-HyRE

For each question, the first model call writes a tentative legal issue frame and a hypothetical reasoning passage. The passage is embedded and used as the retrieval query. The second model

Table 1: Coverage of the expected 84-cell answer ladder. Generated-method cells are HyDE, Snap-HyRE, and rewrite.

Dataset	Model	Signed cells	Missing generated cells
BarExamQA	Ministral 8B	4/7	HyDE, Snap-HyRE, Rewrite
BarExamQA	Gemma 4 26B	7/7	NONE
BarExamQA	Llama 3.3 70B	7/7	NONE
HousingQA	Ministral 8B	0/7	HyDE, Snap-HyRE, Rewrite
HousingQA	Gemma 4 26B	0/7	HyDE, Snap-HyRE, Rewrite
HousingQA	Llama 3.3 70B	4/7	HyDE, Snap-HyRE, Rewrite
CaseHOLD	Ministral 8B	0/7	HyDE, Snap-HyRE, Rewrite
CaseHOLD	Gemma 4 26B	1/7	HyDE, Snap-HyRE, Rewrite
CaseHOLD	Llama 3.3 70B	7/7	NONE
SCALR	Ministral 8B	7/7	NONE
SCALR	Gemma 4 26B	7/7	NONE
SCALR	Llama 3.3 70B	7/7	NONE

call receives the original question and the retrieved corpus passages, then emits the final answer. The hypothetical passage is not treated as evidence in the final call. Older logs may use `rag_snap_hyde_2call`; in this paper that is only a legacy alias for the same two-call family.

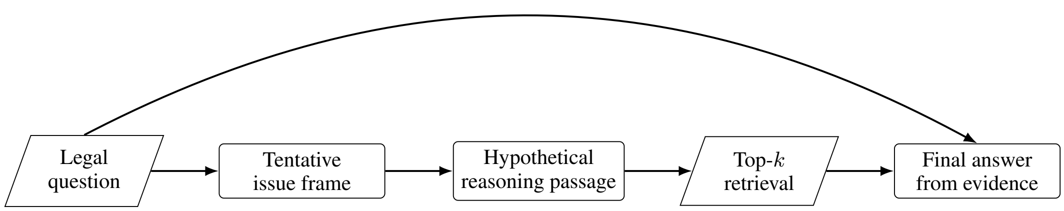

Figure 1: Snap-HyRE retrieval. Generated text shapes retrieval; final answer generation is grounded in the original question and retrieved corpus passages.

2.3 Run Accounting

Conceptually, `rag_hyde`, `snap_hyre`, and `rag_rewrite` are two-call methods when run end-to-end. Some answer rows replay already-built generation and retrieval caches, so answer-package `avg_llm_calls` can show one answer call even for a two-call method. That field is an answer-run accounting statistic, not the conceptual method cost.

2.4 Metrics

The answer metric is task accuracy. Retrieval exposure is exact-qrel Hit@k and MRR@k over labeled gold evidence ids or legal holdings. Exact-qrel metrics are useful but incomplete: they can miss legally equivalent retrieved passages, and they can overstate usefulness when the gold id appears among strong distractors. For that reason, the paper reports retrieval metrics, answer accuracy, oracle evidence controls, and paired McNemar tests together.

This evaluation follows the retrieval-augmented generation and generated-query line of work [Lewis et al., 2020, Gao et al., 2023, Wang et al., 2023], applied to legal benchmarks including LegalBench-SCALR, CaseHOLD, BarExamQA, and HousingQA [Guha et al., 2023, Zheng et al., 2021, 2025].

3 Results

3.1 Answer Grid

Table 2: Current signed full-corpus answer accuracy at $k = 5$ (%). TBD cells are absent from the current fixed-method package rather than inferred from older runs.

Dataset / model	N	LLM	Raw RAG	HyDE	Snap-HyRE	Rewrite	Gold	Gold+Nbrs
BarExamQA / Ministral 8B	1195	56.8	56.9	TBD	TBD	TBD	64.6	63.2
BarExamQA / Gemma 4 26B	1195	80.8	78.0	80.2	82.0	80.7	78.6	80.7
BarExamQA / Llama 3.3 70B	1195	78.7	74.6	80.2	79.8	77.2	79.2	77.8
HousingQA / Ministral 8B	6853	TBD	TBD	TBD	TBD	TBD	TBD	TBD
HousingQA / Gemma 4 26B	6853	TBD	TBD	TBD	TBD	TBD	TBD	TBD
HousingQA / Llama 3.3 70B	6853	44.8	47.3	TBD	TBD	TBD	67.3	66.0
CaseHOLD / Ministral 8B	3600	TBD	TBD	TBD	TBD	TBD	TBD	TBD
CaseHOLD / Gemma 4 26B	3600	72.6	TBD	TBD	TBD	TBD	TBD	TBD
CaseHOLD / Llama 3.3 70B	3600	71.8	70.8	70.3	70.5	70.6	97.5	79.4
SCALR / Ministral 8B	571	67.2	68.0	71.1	69.9	69.9	93.2	77.1
SCALR / Gemma 4 26B	571	73.0	73.4	72.2	73.9	73.9	97.9	81.3
SCALR / Llama 3.3 70B	571	74.4	72.9	70.4	71.3	71.6	93.5	83.0

Table 3: Unbalanced averages across completed clean answer cells. These means summarize the current package only; they are not benchmark-balanced leaderboard scores.

Mode	Cells	Mean acc.	Mean Δ vs raw
LLM-only	9	68.9	0.7
Raw RAG	8	67.7	0.0
HyDE	6	74.1	1.2
Snap-HyRE	6	74.6	1.6
Rewrite	6	74.0	1.1
Gold	8	84.0	16.3
Gold+Nbrs	8	76.1	8.3

The completed-cell means are only an orientation check because the grid is unbalanced. The safer result object is Table 2: BarExamQA has two signed positive Snap-HyRE cells, SCALR has mixed generated-query answer behavior, CaseHOLD has a signed Llama 70B row with little answer conversion, and HousingQA generated-method answer rows remain TBD.

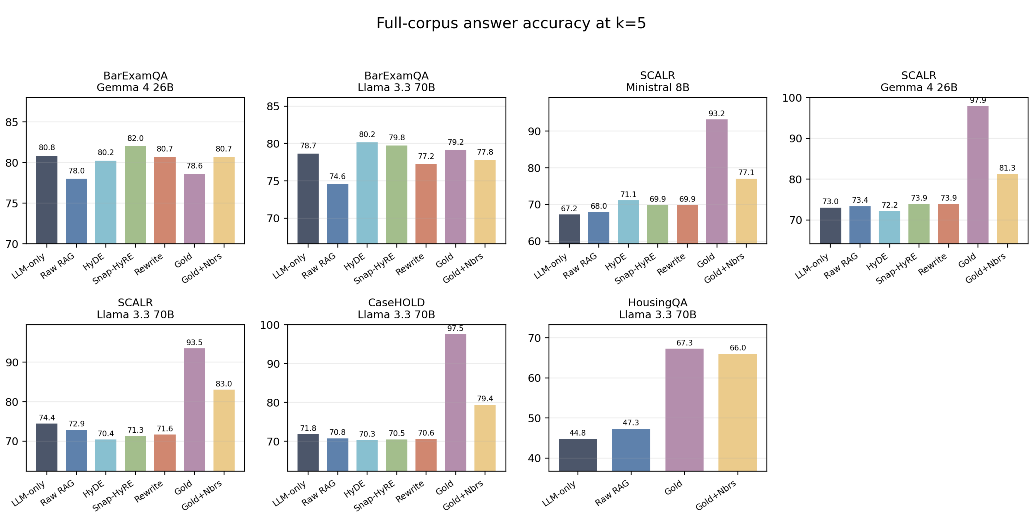

Figure 2: Answer ladders for currently signed cells. This figure is a visual companion to Table 2; missing cells are omitted rather than filled from older runs.

3.2 Retrieval and Answer Conversion

Table 4: Retrieval exposure and answer conversion at $k = 5$ for generated-query cells with signed full answer and retrieval rows. Deltas are method accuracy minus raw-RAG accuracy.

Dataset	Model	Raw Hit	Raw Acc.	HyDE Hit	HyDE Δ	Snap Hit	Snap Δ	Rewrite Hit	Rewrite Δ
BarExamQA	Gemma 4 26B	1.4	78.0	11.4	+2.26	12.1	+4.02	12.2	+2.68
BarExamQA	Llama 3.3 70B	1.4	74.6	10.5	+5.61	11.1	+5.19	12.2	+2.68
SCALR	Ministral 8B	49.6	68.0	60.3	+3.15	62.0	+1.93	65.0	+1.93
SCALR	Gemma 4 26B	49.6	73.4	70.8	-1.23	72.7	+0.53	67.4	+0.53
SCALR	Llama 3.3 70B	49.6	72.9	61.5	-2.45	55.2	-1.57	57.6	-1.22
CaseHOLD	Llama 3.3 70B	17.9	70.8	51.2	-0.42	45.0	-0.25	45.1	-0.14
HousingQA	Llama 3.3 70B	TBD	47.3	TBD	TBD	TBD	TBD	TBD	TBD

Table 4 is the central failure-analysis table. On BarExamQA, generated retrieval handles improve raw retrieval exposure and Snap-HyRE improves answer accuracy. On SCALR and CaseHOLD, higher exact Hit@5 often fails to become higher final accuracy. This is the main reason the paper should avoid a simple retrieval-win narrative.

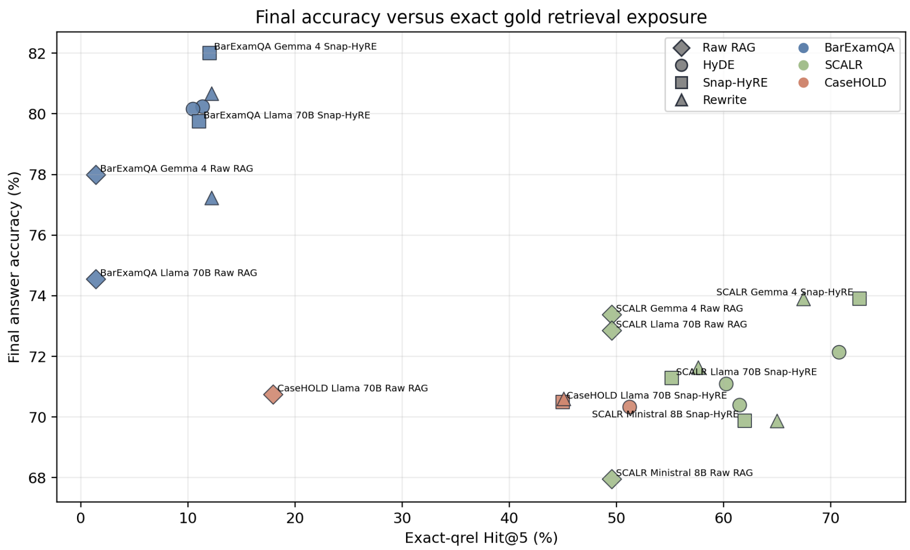

Figure 3: Final answer accuracy versus exact-qrel Hit@5 for signed generated-query cells and their raw-RAG baselines. The same retrieval lift can correspond to positive, flat, or negative answer movement.

Table 5: Paired McNemar comparisons for Snap-HyRE against selected controls. Deltas are percentage points in answer accuracy.

Dataset	Model	Δ raw	p raw	Δ LLM	p LLM	Δ HyDE	p HyDE
BarExamQA	Gemma 4 26B	+4.02	0.000699	+1.17	0.348	+1.76	0.0987
BarExamQA	Llama 70B	+5.19	2.70e-05	+1.09	0.388	-0.42	0.754
SCALR	Ministral 8B	+1.93	0.260	+2.63	0.110	-1.23	0.457
SCALR	Gemma 4 26B	+0.53	0.780	+0.88	0.560	+1.75	0.220
SCALR	Llama 70B	-1.58	0.281	-3.15	0.0222	+0.88	0.542
CaseHOLD	Llama 70B	-0.25	0.722	-1.31	0.0295	+0.17	0.812

3.3 Oracle Evidence Controls

Oracle rows separate retrieval failure from evidence-use failure. Gold-only accuracy is very high on SCALR and CaseHOLD, but gold-plus-neighbors is lower than gold-only, showing that extra plausible legal text can dilute the signal. HousingQA also has a large oracle gap: Llama 70B raw RAG is 47.3%, while gold-only is 67.3%.

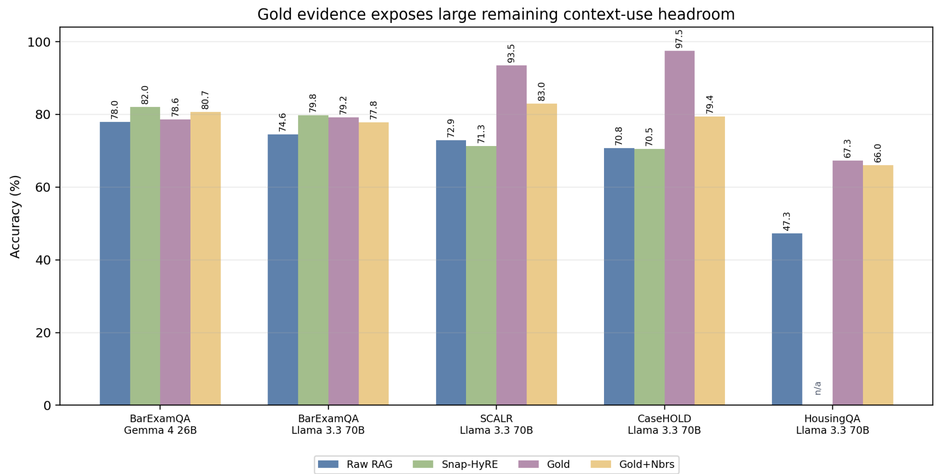

Figure 4: Oracle evidence controls expose remaining context-use headroom. Gold-plus-neighbor rows should not be read as an oracle ceiling; they also test distractor sensitivity.

3.4 Top-k Analyses

The main answer grid uses $k = 5$. Retrieval curves through $k = 10$ are still reported because they show whether generated-query methods move gold evidence into the candidate set. The q100 prelaunch probe is not a final top-k benchmark; it only justifies using $k = 5$ as the shared answer setting while retaining k-curves for analysis.

Table 6: q100 prelaunch top-k probe. Retrieval exposure rises after $k = 5$, but MRR changes little and BarExamQA downstream accuracy did not favor $k = 10$.

k	Macro Hit@k	Macro MRR@k	BarExam q100 acc.
1	19.5	0.195	N/A
5	31.7	0.239	raw 83, HyDE 87
10	36.2	0.245	raw 81, HyDE 84

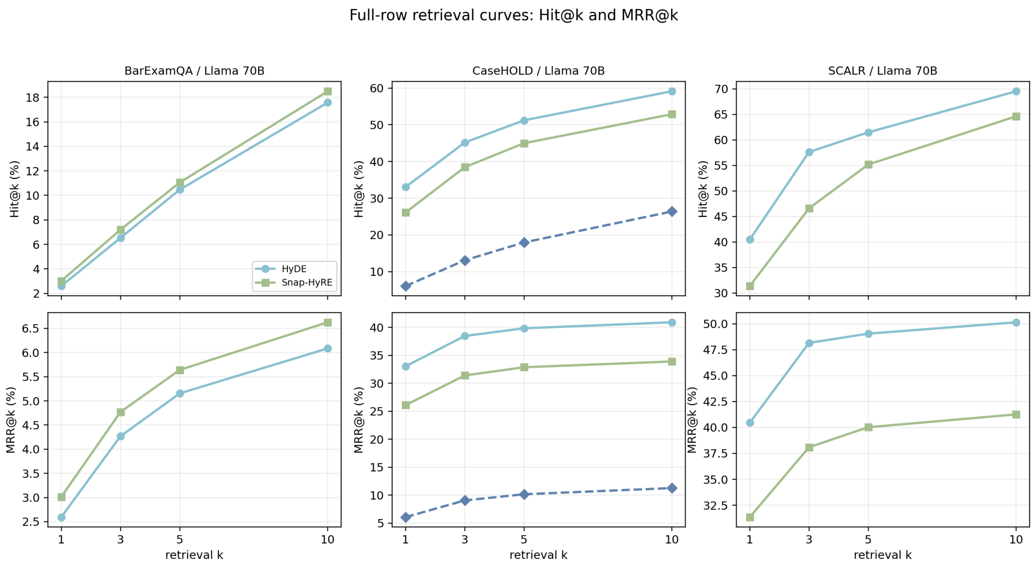

Figure 5: Full-row retrieval curves for signed caches with at least 500 rows. Hit@k and MRR@k are shown separately; raw full curves are available in this package for CaseHOLD and are otherwise summarized at k=5 in Table 4.

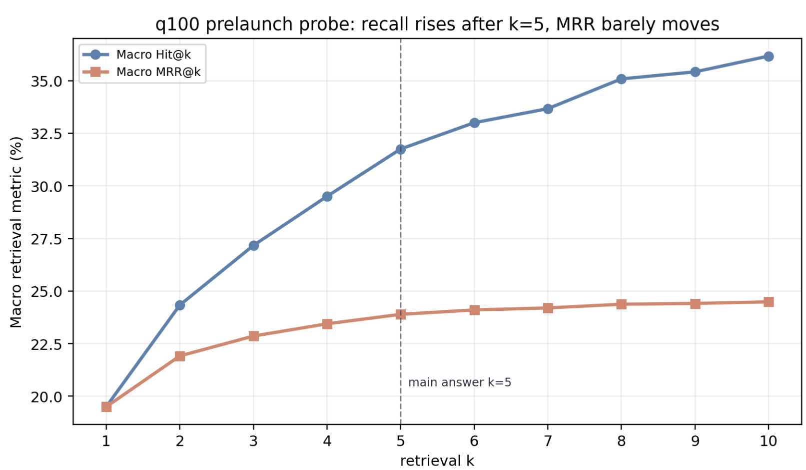

Figure 6: q100 prelaunch retrieval probe over supplied Gemma 4 26B caches. Macro Hit@k rises after k=5, while MRR changes little; the small downstream BarExamQA check did not favor k=10.

3.5 Row Caveats

Table 7: Caveats that should travel with exact row claims. Detailed provenance remains in the signoff log.

Dataset	Model	Row	Caveat
BarExamQA	Gemma 4 26B	Snap-HyRE	OpenRouter upstream retries recovered in place; 3 answer-format retries; 4 near-cap rows.
BarExamQA	Llama 70B	Snap-HyRE	Clean row with one final-answer repair.
SCALR	Ministral 8B	Snap-HyRE	Retry caveat; 9 final-answer repairs; no fallback keys.
SCALR	Gemma 4 26B	Snap-HyRE	10 answer-format retries; 5 near-cap pre-repair rows.
SCALR	Llama 70B	Snap-HyRE	Clean signed row.
CaseHOLD	Llama 70B	Snap-HyRE	Mixed same-model provider recovery and repaired cache row; 16 answer-format retries.
HousingQA	All	Generated methods	Full generated-method answer rows are TBD in the current package.

4 Current Reading

The current evidence supports three narrow statements. First, Snap-HyRE is a real BarExamQA answer improvement over raw RAG in two signed full-corpus model cells. Second, retrieval exposure is not a sufficient proxy for legal answer quality: SCALR and CaseHOLD have generated-query retrieval gains without reliable answer gains. Third, oracle controls remain essential because they show whether the final model can use isolated gold evidence and how much performance is lost when that evidence is mixed with neighboring legal text.

5 Limitations and TBD

The grid is incomplete. The headline missing cells are HousingQA generated methods, CaseHOLD generated methods for Gemma 4 26B and Ministral 8B, and BarExamQA generated methods for Ministral 8B. The current package has 51/84 expected answer cells, so averages across completed cells are unbalanced.

The retrieval metrics are exact-qrel metrics. They should be kept, but they should not be the only mechanism claim. Exact gold ids can miss useful equivalent evidence, while a hit can be surrounded by distractors that make the answer harder. The oracle and gold-plus-neighbor rows are the current guardrail against over-reading Hit@k.

The top-k evidence is also deliberately limited. The q100 prelaunch probe says $k = 10$ raises retrieval exposure but does not justify changing the main answer grid from $k = 5$. It should be cited as a launch gate and analysis prompt, not as a global top-k optimum.

6 Conclusion

Snap-HyRE is best presented as a fixed generated-query legal-RAG method with mixed but informative results. It improves BarExamQA in the current signed cells, does not reliably convert larger retrieval exposure into better answers on SCALR or CaseHOLD, and still lacks generated-method answer rows on HousingQA. The paper's durable contribution is the evaluation pattern: answer accuracy, retrieval exposure, oracle evidence controls, coverage, and row caveats should travel together.

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